

Notes: Jovanovic (JPE, 1979)

Job Matching and the Theory of Turnover

Contents

1	Big Picture	3
2	Incremental Contribution of the Paper	3
3	Economic Setting and Model Objects	4
3.1	Environment	4
3.2	Beliefs and Bayesian Updating	4
4	Equilibrium Wage Contract	4
4.1	Wages as Conditional Expected Marginal Product	4
4.2	Why the Contract Supports Efficient Turnover	5
5	Optimal Stopping, Reservation Wages, and Turnover	5
5.1	State Variables and Wage Process	5
5.2	Reservation Wage	5
6	Predictions	6
6.1	Prediction 1: Separation Probability Falls with Tenure	6
6.2	Prediction 2: Average Wages Rise with Tenure	6
6.3	Prediction 3: Turnover Declines over the Life Cycle	6
7	Interpretation and Connection to Empirical Evidence	6
7.1	What the Model Explains Well	6
7.2	What the Model Abstracts From	7
8	How to Read the Paper for Research Use	7
8.1	The Theoretical Mechanism	7
8.2	Conceptual Contribution for Modern Work	7
9	Key Equations and Propositions from the Paper	8
9.1	Extracts 1 and 2: Title abstract and model output process / posterior updating . . .	8

9.2	Extracts 3 and 4: Equilibrium wage contract and optimal stopping setup	9
9.3	Extracts 5 and 6: Reservation wage, separation density, and wage-tenure profile . . .	10
9.4	Extract 7: Renewal equation and life-cycle turnover	11
10	Conclusion	11

1 Big Picture

One-sentence summary. This note summarizes Jovanovic (1979), which develops a continuous-time equilibrium model of job turnover in which match quality is unknown *ex ante*, workers and firms learn about match-specific productivity from output realizations, wages equal conditional expected productivity in equilibrium, and separations occur through an optimal stopping rule, generating declining turnover with tenure and rising wages among surviving matches.

Core question. Why do permanent job separations decline sharply with job tenure, and why do wages tend to rise with tenure and labor market experience, even without assuming that all wage growth comes from human capital accumulation?

Main answer. A job is treated as an *experience good*: the quality of a worker-employer match cannot be known before the match is tried. As output is observed over time, beliefs about match quality become more precise. Bad matches tend to end early; good matches survive. This selection and learning mechanism produces the key empirical patterns of turnover and wage dynamics.

Central intuition. The model is not primarily about a worker's fixed ability. Instead, it is about the quality of a particular match. A worker may be productive at one firm or task and unproductive at another. Tenure reveals whether the current match is good. Longer-tenure matches are therefore positively selected survivors.

2 Incremental Contribution of the Paper

- **Equilibrium theory of turnover.** The paper provides one of the first explicit equilibrium treatments of permanent job turnover generated by learning about current match quality.
- **Job as an experience good.** It formalizes the idea that a job's quality is learned only after the worker and employer form the match, distinguishing this mechanism from pure search-good models where workers leave because they find new outside options.
- **Wage contract result.** The paper proves that paying each worker his conditional expected marginal product is an equilibrium wage contract.
- **Turnover uniqueness.** Even though the exact wage contract may not be unique, the induced turnover behavior is unique in equilibrium.
- **Tenure-wage and tenure-turnover implications.** The same learning process explains both declining separations with tenure and increasing average wages among workers who remain on the job.
- **Selection rather than only accumulation.** The model offers an alternative to pure human-capital explanations of wage growth: wages rise with tenure partly because low-match-quality workers exit and high-match-quality workers stay.

3 Economic Setting and Model Objects

3.1 Environment

The economy has competitive firms with constant returns to scale and labor as the only input. A worker's output contribution in a match is stochastic. The worker and the employer both observe output, so the model avoids asymmetric information between the current employer and the worker.

The core output process is

$$X(t) = \mu t + \sigma z(t),$$

where $X(t)$ is cumulative output over tenure t , $z(t)$ is a standard Wiener process, and μ is the unknown productivity of the worker in the current match.

Interpretation of μ . The parameter μ is not general worker ability. It is match quality. A worker who separates draws a new μ from the population distribution of match qualities. Successive draws are independent, so information learned in one match is match-specific.

3.2 Beliefs and Bayesian Updating

At the start of a match, μ is unknown. The prior distribution is normal with mean m and variance s . After observing tenure t and cumulative output $X(t) = x$, the posterior distribution of μ is normal, with posterior mean

$$E_{x,t}(\mu) = (ms^{-1} + x\sigma^{-2}) (s^{-1} + t\sigma^{-2})^{-1}$$

and posterior variance

$$S(t) = (s^{-1} + t\sigma^{-2})^{-1}.$$

The pair $[X(t), t]$ is therefore a sufficient statistic for everything learned about the match. As tenure rises, posterior variance falls; the worker and firm become increasingly certain about the match.

4 Equilibrium Wage Contract

4.1 Wages as Conditional Expected Marginal Product

Let

$$w^*(x, t) = E_{x,t}(\mu).$$

This wage rule pays the worker the conditional expected productivity of the current match at every moment. Jovanovic shows that this contract is an equilibrium.

Economic meaning. In a competitive labor market, firms bid for workers. Since firms are risk neutral and information is symmetric, the worker receives the expected marginal product of the match. The current employer cannot systematically underpay a worker relative to the value of the information already generated by the match.

4.2 Why the Contract Supports Efficient Turnover

The worker quits when the value of continuing the current match falls to the value of starting over, net of job-changing costs. The firm can induce separations by lowering the wage enough to make quitting optimal; empirically, these separations correspond to total separations, including both quits and disguised layoffs.

The equilibrium has three key conditions:

1. Workers follow the optimal quitting policy given the wage contract.
2. Firms maximize expected profits given worker behavior.
3. Expected profits are zero under competition.

The model's equilibrium turnover behavior is Pareto efficient given the information available: a match ends exactly when the expected rent from the match falls to zero.

5 Optimal Stopping, Reservation Wages, and Turnover

5.1 State Variables and Wage Process

The relevant state variables can be written as the current wage estimate $w(t)$ and tenure t , where

$$w(t) = E_{X(t),t}(\mu).$$

The wage follows a driftless process:

$$dw(t) = S(t)\sigma^{-1}dz(t), \quad w(0) = m.$$

The key feature is that the incremental variance of wages declines over time because $S(t)$ declines with tenure. Early in the match, new output observations are highly informative; later, the posterior is already precise, so each new signal moves beliefs less.

5.2 Reservation Wage

Let Q be the present value of quitting and taking the best alternative. There is a reservation wage boundary $\theta(t)$ such that the worker stays when $w(t) > \theta(t)$ and quits when $w(t) \leq \theta(t)$. The boundary satisfies

$$\theta(t) = rQ - \frac{S(t)^2}{2\sigma^2} V_{ww}[\theta(t), t].$$

Jovanovic shows that

$$\theta(t) < rQ, \quad \frac{d\theta(t)}{dt} \geq 0, \quad \lim_{t \rightarrow \infty} \theta(t) = rQ.$$

Economic meaning. Early in the match, uncertainty has option value. Even if current expected productivity is somewhat low, the worker may stay because future information may reveal a good match. As tenure rises, less uncertainty remains, so the option value of waiting falls and the reservation wage rises toward rQ .

6 Predictions

6.1 Prediction 1: Separation Probability Falls with Tenure

The central prediction is that turnover eventually declines with tenure. Bad matches tend to be discovered early because early output signals are informative. Workers who survive longer are more likely to have good matches, and there is less new information left to overturn that conclusion.

The paper derives the separation distribution from a first-passage problem. Using an approximation in which the reservation wage is set equal to rQ , the model obtains a closed-form density for separations by tenure. The separation hazard is nonmonotonic in theory: it may initially rise, but it must eventually decline to zero. Empirically, if the mode is close to zero, the hazard declines early, matching the observed negative tenure-turnover relationship.

6.2 Prediction 2: Average Wages Rise with Tenure

Surviving workers are positively selected: low-wage, low-match-quality workers quit; high-wage, high-match-quality workers stay. Therefore, the average wage among workers who remain with a job rises with tenure and eventually flattens out.

This is an important alternative to a pure on-the-job human capital explanation. In this model, wage growth can occur because of learning and selection, even when the worker's underlying match productivity μ is fixed.

6.3 Prediction 3: Turnover Declines over the Life Cycle

Because job tenure and labor market experience are correlated, declining turnover with tenure can also generate declining turnover over the life cycle. The model treats job durations as independent draws from a renewal process and derives a renewal equation for the labor-market-experience separation density.

7 Interpretation and Connection to Empirical Evidence

7.1 What the Model Explains Well

- **Strong tenure-turnover gradient.** Turnover falls as tenure rises because information about the match becomes more precise.
- **Wage-tenure growth.** Wages rise with tenure because remaining matches are better matches on average.
- **Mover-stayer patterns.** Workers with many separations are more likely to have experienced poor matches and therefore lower average past wages, holding other factors fixed.
- **Symmetry of quits and layoffs.** The paper can rationalize both quits and layoffs as separations generated by low revealed match quality, since the firm can lower wages to induce the worker to leave.

7.2 What the Model Abstracts From

- It abstracts from aggregate demand shocks and temporary layoffs.
- It assumes symmetric information between the worker and the employer about current output.
- It assumes that match quality draws across jobs are independent.
- It does not explicitly model strategic bargaining or asymmetric outside offers.
- It treats the job as an experience good rather than an inspection good.

8 How to Read the Paper for Research Use

8.1 The Theoretical Mechanism

The paper is useful whenever a research question involves selection through learning. The structure is:

unknown match quality $\rightarrow$ signals from performance

$\rightarrow$ posterior updating

$\rightarrow$ stopping decision

$\rightarrow$ observed selection in outcomes.

The model can be reused in contexts beyond labor turnover, including client-firm relationships, bank-borrower matches, supplier relationships, venture capital screening, or advisor-client relationships, whenever continuation depends on information accumulated inside a relationship.

8.2 Conceptual Contribution for Modern Work

The paper separates two ideas often conflated in empirical labor and organizational studies:

1. **True productivity growth:** workers become more productive because they accumulate human capital.
2. **Selection through learning:** observed workers look more productive because bad matches disappear and good matches remain.

For empirical work, this distinction matters because a positive tenure-wage slope need not imply that tenure itself causes productivity growth. It may partly reflect learning about match quality.

9 Key Equations and Propositions from the Paper

9.1 Extracts 1 and 2: Title abstract and model output process / posterior updating

Job Matching and the Theory of Turnover

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A long-run equilibrium theory of turnover is presented and is shown to explain the important regularities that have been observed by empirical investigators. A worker's productivity in a particular job is not known ex ante and becomes known more precisely as the worker's job tenure increases. Turnover is generated by the existence of a nondegenerate distribution of the worker's productivity across different jobs. The nondegeneracy is caused by the assumed variation in the quality of the worker-employer match.

Extract 1: Title, author, and abstract.

JOB MATCHING AND TURNOVER

313

The Model

Assume that firms' production functions exhibit constant returns to scale and that labor is the only factor of production. Under competitive conditions, the size of firm is then indeterminate. Each worker's output is assumed to be observed instantaneously by the worker and by the employer so that informational asymmetries do not arise. Let $X(t)$ be the contribution by a worker to the total output of the firm over a period of length t , and let

$$X(t) = \mu t + \sigma z(t) \quad (\text{for each } t > 0) \quad (1)$$

where μ and σ are constants and $\sigma > 0$, and where $z(t)$ is a standard normal variable with mean zero and variance t (a standard Wiener process with independent increments so that $\text{cov}[z(t), z(t')] = \min[t, t']$). Then $X(t)$ is normally distributed with mean μt and with variance $\sigma^2 t$. Assume that σ is the same for each firm-worker match while in general μ differs across matches. The interpretation of μ is not one of worker ability but a measure of the quality of the match. When the match is formed, μ is unknown. As the match continues, further information (in the form of output as given by eq. [1]) is generated. A "good match" is one possessing a large μ . Let μ be normally distributed across matches, with mean m and with variance s , and assume that job changing involves drawing a new value of μ from this distribution and the successive drawings are independent. The latter assumption guarantees that the worker's prior history is of no relevance in assessing his μ on a newly formed match. The only way to learn about μ is to observe the worker on the job for a period of time. This independence assumption also means that the informational capital thus generated is completely match specific and is analogous to the concept of firm-specific human capital.²

For a worker with job tenure t and cumulative output $X(t) = x$ the above assumptions imply that the available information on μ on his current job can be characterized by a posterior distribution that is normal (see Chernoff 1968, p. 266) with

$$\begin{aligned} \text{posterior mean} &\equiv E_{xt}(\mu) = (ms^{-1} + x\sigma^{-2})(s^{-1} + t\sigma^{-2})^{-1} \\ \text{posterior variance} &\equiv S(t) = (s^{-1} + t\sigma^{-2})^{-1}. \end{aligned} \quad (2)$$

The pair $[X(t), t]$ is therefore a sufficient statistic for the information contained in the entire posterior distribution. (This property is essentially due to the independent increments property of the Wiener

Extract 2: Output process and posterior learning.

The paper begins by defining turnover as learning about match quality and then formalizes match output as a noisy signal of μ .

Read. The paper's abstract gives the entire logic: productivity in a job is unknown ex ante and becomes more precise with tenure. Extract 2 shows the formal law of motion for output and the Bayesian posterior formulas. The posterior variance $S(t)$ declines as tenure grows, which is the mathematical engine behind the declining value of new information.

Economic message. Turnover is not a random event layered on top of wage dynamics. Turnover is the optimal response to learning about a specific relationship.

9.2 Extracts 3 and 4: Equilibrium wage contract and optimal stopping setup

910

JOURNAL OF POLITICAL ECONOMY

policy in response to $w(\cdot)$ and to $Q([w])$; (E2) $\pi\{Q([w]), [w]\} \geq \pi\{Q([w]), [\hat{w}]\}$ for all $\hat{w}(\cdot) \neq w(\cdot)$, so that $w(\cdot)$ maximizes expected profits; (E3) $\pi\{Q([w]), [w]\} = 0$ (zero expected profit constraint). Let $w^*(x, t) = E_{x,t}(\mu)$ for all (x, t) . This wage contract states that the worker will be paid his expected (marginal) product at each moment in time. Let $Q^* = Q([w^*])$.

Theorem 1.— $w^* \in B$.

Proof.—E3 is clearly satisfied by w^* . To prove E1 and E2, suppose by contradiction that E2 is not satisfied by w^* so that there exists some $w \in B$ such that a deviant firm offers it

$$\pi(Q^*, [w^*]) < \pi(Q^*, [w]) \quad (6)$$

while the worker must be doing at least as well as under w^* :

$$\alpha(Q^*, [w^*]) \leq \alpha(Q^*, [w]). \quad (7)$$

(The value of quitting the deviant firm is unchanged at Q^* .) From (5),

$$\begin{aligned} \pi(Q^*, [w]) - \pi(Q^*, [w^*]) + \alpha(Q^*, [w]) - \alpha(Q^*, [w^*]) &= \beta(Q^*, [w]) \\ &- \beta(Q^*, [w^*]) + \gamma(Q^*, [w]) - \gamma(Q^*, [w^*]). \end{aligned} \quad (8)$$

Then equations (6) and (7) imply that the left-hand side of (8) is strictly positive. But the right-hand side of (8) is equal to $\int_0^\infty e^{-rt} \int_{-\infty}^\infty w^*(x, t) \{h(x, t | [w], Q^*) - h(x, t | [w^*], Q^*)\} dx dt + Q^* \int_0^\infty e^{-rt} \{f(t | [w], Q^*) - f(t | [w^*], Q^*)\} dt$, and this expression cannot be positive since the quitting policy implied by $\{h(x, t | [w^*], Q^*), f(t | [w^*], Q^*)\}$ is optimal for the workers when faced with the wage contract $w^*(x, t)$ and the present value of quitting Q^* . *Q.E.D.*

Since workers and firms are risk neutral, $w^*(x, t)$ is not a unique equilibrium contract, any random variable ξ possessing the property $E_{x,t}(\xi) = w^*(x, t)$ would also qualify. A pure piece-rate wage involving a payment of $X(t + \Delta t) - X(t)$ over the interval $(t, t + \Delta t)$ therefore also qualifies as equilibrium since $E_{x,t}dX(t) = w^*(x, t)dt + \sigma E_{x,t}dz(t) = w^*(x, t)dt$. Any such contract leads to identical turnover behavior as under $w^*(x, t)$. Even within the class of functions of x and t alone, $w^*(x, t)$ may not be unique. The following theorem guarantees, however, that turnover behavior is unique.

Theorem 2.—If $w \in B$ then $h\{x, t | [w], Q([w])\} = h\{x, t | [w^*], Q([w^*])\}$, and $f\{t | [w], Q([w])\} = f\{t | [w^*], Q([w^*])\}$.

Proof.—See Jovanovic 1978a. The proof is lengthy and not particularly instructive. Theorem 2 states that the separation policy of the worker is unique even though the wage contract leading to it is not. This turnover behavior is identical with that which results in a situation in which each firm offers a wage contract $w^*(x, t) = E_{x,t}(\mu)$.

Extract 3: Equilibrium wage contract and turnover uniqueness.

The equilibrium contract pays conditional expected productivity, and the worker's continuation decision is an optimal stopping problem.

Read. Extract 3 states that the wage $w^*(x, t) = E_{x,t}(\mu)$ is an equilibrium and that the resulting turnover behavior is unique. Extract 4 rewrites the problem in terms of a stochastic wage process with declining variance and derives the Bellman equation governing continuation.

Economic message. The model makes turnover endogenous to the information process. Wages are not just prices; they also summarize the posterior belief about the match.

knew *which* workers and *which* firms would make good matches.

Assume that the worker is faced with the wage contract $w^*(x, t) = E_{x,t}(\mu)$ and a present value of quitting Q . The sufficient statistics (state variables) are $X(t)$ and t . It is more convenient to use instead $w(t)$ and t as the two state variables, where $w(t) = E_{x(t)}(\mu)$. Since $w(t)$ is normally distributed with mean m and variance $s - S(t)$ for all t , it satisfies the stochastic differential equation

$$dw(t) = S(t)\sigma^{-1}dz(t) \quad w(0) = m \quad (9)$$

so that the worker's wage follows a driftless random process with ever-decreasing incremental variance that tends to zero as tenure tends to infinity. Let $V(w, t)$ be the ("current") value of the game to the worker who has tenure t and wage $w(t) = w$. Then letting $E_{w,t}$ denote the mathematical expectation operator conditioned upon w and t ,⁶

$$V(w, t) = w\Delta t + e^{-r\Delta t}E_{w,t}V(w[t + \Delta t], t) + o(\Delta t). \quad (10)$$

Subtracting $V(w, t)$ from both sides, dividing through by Δt , taking the limit as Δt tends to zero, and applying Itô's Lemma (see Kushner 1971) yields

$$w - rV(w, t) + \frac{S(t)^2}{2\sigma^2}V_{ww}(w, t) + V_t(w, t) = 0. \quad (11)$$

As with most optimal stopping problems involving Markov processes, the space of points (w, t) can be divided into a continuation region and a stopping region (see Shiryaev 1973). The continuation region consists of those wage-tenure combinations at which it is optimal for the worker to remain with the firm. Equations (10) and (11) hold for all

⁶ $o(\Delta t)$ represents terms tending to zero faster than Δt does. Note that the option of stopping on $(t, t + \Delta t)$ (in which case a reward Q is collected) is exercised with a probability that behaves essentially as does

Extract 4: Wage process and Bellman equation.

9.3 Extracts 5 and 6: Reservation wage, separation density, and wage-tenure profile

980

JOURNAL OF POLITICAL ECONOMY

wage-tenure combinations that belong to the continuation region. Let $[\theta(t), t]$ be the boundary of the continuation region so that along the boundary $V[\theta(t), t] = Q$, and $\theta(t)$ may be thought of as the reservation wage at which the worker quits the firm. Evaluating equation (11) at $w = \theta(t)$, $\theta(t) = rQ - [S(t)^2/2\sigma^2]V_{ww}[\theta(t), t] - V_t[\theta(t), t]$. A well-known "smooth-fit" condition of optimal stopping (see Shiryaev 1973) states that along the boundary,⁷ $V_t[\theta(t), t] = \partial Q/\partial t = 0$, implying that

$$\theta(t) = rQ - \frac{S(t)^2}{2\sigma^2} V_{ww}[\theta(t), t]. \quad (12)$$

In the interior of the continuation region $V(w, t) > Q$. Since at the reservation wage $V[\theta(t), t] = Q$,⁸ and since $V_w[\theta(t), t] = 0$, this implies that $V_{ww}[\theta(t), t] \geq 0$. Note that $S(t)$ declines monotonically to zero which suggests that $\theta(t)$ should be monotonically increasing up to rQ . It is possible to prove (see the Appendix) that $\theta(t) < rQ$ for all t , that $d\theta/dt \geq 0$, and that $\lim_{t \rightarrow \infty} \theta(t) = rQ$ so that the reservation wage increases up to its limit from below. The reason for the increase in the reservation wage is the decrease of the incremental variance of the wage process as tenure increases. A large incremental variance implies a large dispersion in possible future wages. If wages turn out to be very high the worker does not quit. If they become very low, the worker partially avoids this adverse outcome by quitting and collecting Q . In the absence of the opportunity to quit, the risk-neutral worker's welfare would be unaffected by changes in the incremental variance. The limit of the reservation wage is rQ . This is because the wage tends to a constant as t tends to infinity. There is nothing further to be learned, and at the point of indifference between staying and quitting the capitalized value of this constant wage must be equal to the present value of quitting, Q .

To obtain an approximation to the probability of job separation by

Extract 5: Reservation wage boundary.

These blocks derive the separation rule and the model's key empirical implications for turnover and wages.

Read. Extract 5 shows that the reservation wage rises toward rQ because uncertainty declines. Extract 6 derives the separation density and explains why the separation hazard eventually declines; it also states the model's wage-tenure implication.

Economic message. Early tenure is a learning-intensive period. Low-quality matches are filtered out early, so long-tenure workers are selected survivors with higher expected match quality.

JOB MATCHING AND TURNOVER

981

$$F(t | [w], Q) = 2 \left[1 - N \left[\frac{m - rQ}{p(t)^{1/2}} \right] \right] = 2N \left[\frac{-m + rQ}{p(t)^{1/2}} \right]$$

with density

$$f(t | [w], Q) = \frac{(2\pi)^{-1/2}(m - rQ)}{p(t)^{3/2}} \exp \left[-\frac{(m - rQ)^2}{2p(t)} \right]$$

where $N(x) = (2\pi)^{-1/2} \int_{-\infty}^x e^{-z^2/2} dz$ where $p(t) = s - S(t)$ is the precision. The unique mode of this distribution is $(m - rQ)^2$. After the mode, the probability of turnover declines rapidly to zero. Some workers never change jobs, since $\lim_{t \rightarrow \infty} F(t | \cdot) < 1$.

To determine the predicted behavior of the separation probability by tenure, consider the hazard rate, $\phi(t) \equiv f/(1 - F)$. Then $\phi(t)$ is the density of separation conditional upon an attained level of tenure, t . The model predicts a nonmonotonic relationship: first $[\phi'(t)] > 0$ and then $\phi'(t) < 0$ as t gets relatively large. That $\phi(t)$ must eventually decline follows since $\lim_{t \rightarrow \infty} f(t) = 0$, while $1 - F(t)$ is bounded away from zero. The precise mathematical expression for the tenure level t^* at which $\phi'(t)$ changes sign and finally becomes negative cannot be obtained in closed form, but since $f' > 0$ implies $\phi' > 0$ clearly $t^* \geq m - rQ =$ the mode of f . If the mode of f is close to zero, $\phi'(t)$ is likely to become negative early on, as appears to be the case empirically (see Jovanovic and Mincer 1978).

The tenure-wage profile (defined as the conditional expectation of the wage given that the worker has attained tenure t) may also be calculated¹⁰ and is equal to $\hat{w}(t) = (m + (m - rQ)2N\{-a[s - S(t)]^{-1/2}\}/1 - 2N\{-a[s - S(t)]^{1/2}\})$. Note that $\hat{w}(t)$ increases monotonically from m when tenure is zero up to $[m + (m - rQ)2N(-as^{-1/2})/1 - 2N(-as^{-1/2})]$ when tenure tends to infinity. Therefore, as low-wage workers quit and high-wage workers stay, the model implies that the average wage of a cohort of workers increases with tenure, eventually at a decreasing rate. In the limit, as tenure becomes indefinitely large, the average wage of those members of the cohort who have not quit approaches a constant as the wage of each worker becomes constant and equal to his true productivity. This then is an alternative explanation for wage

Extract 6: Turnover density, hazard, and wage-tenure profile.

9.4 Extract 7: Renewal equation and life-cycle turnover

982

JOURNAL OF POLITICAL ECONOMY

A mismatch leads to a lower wage and an early separation. Thus, holding constant market experience, average past earnings are likely to be lower for a worker who has experienced many job separations.¹¹ This prediction appears to be consistent with evidence from the National Longitudinal Study (NLS) mature men's sample (see Bartel and Borjas 1976).

Job durations over the life cycle are identically and independently distributed random variables. The turnover generated by the model therefore forms a pure renewal process (see Feller 1966, chap. 11). Let y denote the worker's labor market experience and $R(y) + o(\Delta y)$ denote the probability that the worker experiences a job separation on the market experience interval $(y, y + \Delta y)$. Then $R(y)$ is the renewal density which satisfies the equation

$$R(y) = f(y) + \int_0^y f(t)R(y - t)dt. \quad (13)$$

Jovanovic and Mincer (1978) prove that a monotonically declining $\phi(t)$ implies a monotonically declining $R(y)$. In other words, a monotonically declining separation probability by tenure is *by itself* sufficient to cause turnover to decline monotonically over the life cycle.¹²

Last, the model generalizes straightforwardly to incorporate permanent differences in workers' characteristics such as level of schooling, ability, race, sex, and so on. The parameters of the model (S, m, σ^2, r) can then be regarded as functions of these variables, with each distinct group of workers treated as though they belonged to a distinct market of workers of that type. The entire analysis remains valid so long as informational symmetry between workers and employers is maintained, so that issues of signaling and self-selection are side-

Extract 7: Renewal equation for turnover over labor-market experience.

Read. The final extract links tenure-based separation probabilities to turnover over labor market experience using a renewal process. If the tenure-based separation probability declines monotonically, turnover over the life cycle also declines.

Economic message. The model connects micro-level match learning to aggregate life-cycle patterns: repeated job matches generate a renewal process whose observable turnover rate declines as workers accumulate experience.

10 Conclusion

Jovanovic (1979) provides a canonical theory of turnover based on match-specific learning. The model starts from a simple premise: the quality of a worker-employer match is not known until it is experienced. As output is observed, the posterior belief about match quality becomes more precise. Competitive equilibrium wages equal conditional expected productivity, and workers optimally quit when the continuation value of the current match falls to the value of starting over.

The key result is that one mechanism explains both turnover and wage dynamics. Bad matches end early, so separations decline with tenure. Good matches survive, so the average wage of workers remaining in a job rises with tenure. This gives a powerful alternative interpretation of wage-tenure profiles: they may reflect learning and selection, not only human capital accumulation.

Takeaway. The paper is foundational because it turns labor turnover into a dynamic information and matching problem. Its logic remains useful for modern research on worker-firm matching, firm learning, productivity dispersion, employment dynamics, and any relationship in which agents learn the quality of a match only after forming it.